Reviewer Pack — Minimal p-Adic Hodge Trace and Communication Complexity Rank…

Entry 231171351957 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-06-08 11:47:53 UTC

Minimal p-Adic Hodge Trace and Communication Complexity Rank Variance

Entry ID: `231171351957` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-06-08 11:47:53 UTC

1. Conjecture

Field A (mathematical branch): p-Adic Number Theory **Field B** (complexity object): Communication Complexity

Statement:

> For every instance of the communication complexity problem with n bits, the minimal p-adic Hodge trace of the characteristic polynomial of its adjacency matrix is linearly correlated with its rank variance, such that |τ(H)| = Θ(Var_Rank(n)), where τ(H) is the minimal p-adic Hodge trace and Var_Rank(n) is the rank variance of the adjacency matrix.

Rationale (proposer's reasoning):

> p-adic Hodge theory provides a framework for studying arithmetic properties of complex algebraic varieties, which may reveal hidden structures in communication complexity. The characteristic polynomial captures the algebraic structure of the problem, and its minimal p-adic Hodge trace could be related to the complexity measures of communication complexity.

Taxonomy category: `communication_complexity` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `45a57a619960dfab`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, across 30 random seeds, the correlation coefficient between |τ(H)| and Var_Rank(n) is found to be greater than or equal to 0.8 AND less than or equal to 1.2.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.80	UNCERTAIN	HITS
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `ADJACENT_OK` against 7 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `p-Adic Hodge trace AND communication complexity` - `adjacency matrix characteristic polynomial rank variance IN p-Adic Number Theory` - `linear correlation between p-adic Hodge trace and communication complexity rank variance`

Top relevant hits considered: - [http://arxiv.org/abs/math-ph/0512018v2] On Phase Transitions for P-Adic Potts Model with Competing Interactions on a Cayley Tree - [http://arxiv.org/abs/1908.08424v1] An introduction to p-adic period rings - [http://arxiv.org/abs/2408.00810v3] p-adic Equiangular Lines and p-adic van Lint-Seidel Relative Bound - [http://arxiv.org/abs/0707.3682v2] On the p-adic Beilinson conjecture for number fields - [http://arxiv.org/abs/hep-th/9410058v3] p-Adic description of Higgs mechanism I: p-Adic square root and p-adic light cone - [http://arxiv.org/abs/1701.06857v2] The p-adic Kummer-Leopoldt constant - Normalized p-adic regulator - [http://arxiv.org/abs/1403.8106v1] Recent advances on the log-rank conjecture in communication complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math
from fractions import Fraction

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_instance(n):
        A = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
        for i in range(n):
            A[i][i] = 1
        return A

    def matrix_multiplication(A, B):
        n = len(A)
        C = [[0 for _ in range(n)] for _ in range(n)]
        for i in range(n):
            for j in range(n):
                for k in range(n):
                    C[i][j] += A[i][k] * B[k][j]
        return C

    def gaussian_elimination(A, b):
        n = len(A)
        M = [A[i] + [b[i]] for i in range(n)]
```

```
    for i in range(n):
        max_row = i
        for j in range(i+1, n):
            if abs(M[j][i]) > abs(M[max_row][i]):
                max_row = j
        M[i], M[max_row] = M[max_row], M[i]
        factor = Fraction(M[i][i])
        for j in range(n + 1):
            M[i][j] /= factor
        for j in range(i+1, n):
            factor = Fraction(M[j][i])
            for k in range(n + 1):
                M[j][k] -= factor * M[i][k]
    x = [0 for _ in range(n)]
    for i in range(n-1, -1, -1):
        x[i] = M[i][n]
        for j in range(i+1, n):
            x[i] -= M[i][j] * x[j]
    return x

def characteristic_polynomial(A):
    n = len(A)
    if n == 1:
        return [A[0][0], -1]
    else:
        det = Fraction(0)
        for j in range(n):
            submatrix = [[A[i][k] for k in range(n) if k != j] for i in range(1, n)]
            det += (-1)**j * A[0][j] * characteristic_polynomial(submatrix)[0]
        return [det, -1]

def minimal_p_adic_hodge_trace(poly):
    return abs(poly[0])

def rank_variance(A):
    n = len(A)
    I = [[int(i == j) for j in range(n)] for i in range(n)]
    diff = matrix_multiplication(A, A) - 2 * A + I
    _, _, det = gaussian_elimination(diff, [0] * n)
    return abs(det)

def correlation(x, y):
    if len(x) != len(y):
        raise ValueError("x and y must have the same length")
    n = len(x)
    mean_x = sum(x) / n
    mean_y = sum(y) / n
    cov = sum((x[i] - mean_x) * (y[i] - mean_y) for i in range(n)) / n
    std_x = math.sqrt(sum((x[i] - mean_x)**2 for i in range(n)) / n)
    std_y = math.sqrt(sum((y[i] - mean_y)**2 for i in range(n)) / n)
    return cov / (std_x * std_y) if std_x != 0 and std_y != 0 else None

results = []
for n in [5, 10, 15, 20, 30, 40]:
    instances_tested = 0
    tau_values = []
```

```
        var_rank_values = []
        for _ in range(5):
            A = generate_instance(n)
            poly = characteristic_polynomial(A)
            tau = minimal_p_adic_hodge_trace(poly)
            var_rank = rank_variance(A)
            if tau is not None and var_rank is not None:
                tau_values.append(tau)
                var_rank_values.append(var_rank)
                instances_tested += 1
        if instances_tested > 0:
            corr = correlation(tau_values, var_rank_values)
            results.append({
                "metric_name": "Correlation",
                "metric_value": corr,
                "instances_tested": instances_tested,
                "n_max": n,
                "conjecture_holds": corr is not None and 0.8 <= abs(corr) <= 1.2,
                "counterexample": ""
            })

    return {
        "metric_name": "Correlation",
        "metric_value": sum(r["metric_value"] for r in results if r["metric_value"] is not None),
        "instances_tested": sum(r["instances_tested"] for r in results),
        "n_max": max(r["n_max"] for r in results),
        "conjecture_holds": all(0.8 <= abs(r["metric_value"]) <= 1.2 for r in results if r["metric
        "counterexample": ""
    }

if __name__ == "__main__":
    seeds = [int(x) for x in sys.argv[1:]] or [2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43,
    results = [run_trial(seed) for seed in seeds]

    mean_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len([r
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    print(f"RESULT: SUPPORTED mean={mean_value} std={math.sqrt(sum((r['metric_value'] - mean_value
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
, 'counterexample': "'Fraction' object is not subscriptable"}
TRIAL: {'metric_name': 'Correlation', 'metric_value': None, 'instances_tested': 0, 'n_max': 5, 'conje
TRIAL: {'metric_name': 'Correlation', 'metric_value': None, 'instances_tested': 0, 'n_max': 5, 'conje
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TRIAL: {'metric_name': 'Correlation', 'metric_value': None, 'instances_tested': 0, 'n_max': 5, 'conje

--STDERR--
```

```
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_d836a386.py", line 120, in <module>
    mean_value = sum(r["metric_value"] for r in results if r["metric_value"] is not None) / len([r for r
                 ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~~~~~~
ZeroDivisionError: division by zero
```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

> The test code crashed before producing data that could confirm or falsify the conjecture. | next: Investigate and fix the error in the test code to proceed with the verification of the conjecture.

11. Audit log (LLM calls)

Total LLM calls: 10

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13116
2	propose	ollama_remote	glm4:latest	0	0	13196
3	preregistration	ollama_remote	glm4:latest	0	0	10862
4	novelty	ollama_remote	glm4:latest	0	0	12401
5	novelty	ollama_remote	glm4:latest	0	0	9491
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	46346
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12479
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14740
9	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16645
10	judge	ollama_remote	glm4:latest	0	0	20309

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 169585 ms total latency. Provider mix: {'ollama_remote': 10}

(full prompt+response transcripts available in `research/audit/231171351957.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/231171351957.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/231171351957.tar.gz` (if generated)